Progress Report 7

Developing a Low-Cost, Machine Learning-Based Early Earthquake Detection System

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Objectives: Prepare for SEAS, VJAS, and GS@IP Research Symposium by validating the prototypes performance by independently testing sensors. Improving the CNN model to achieve stronger results.

For the testing process, we had originally discussed different options for obtaining a shake simulation table. That included contacting several labs in the area, universities, etc. We had gotten a response from George Mason University, although their shake table was severely bugged and needed to be fixed, crossing them out of the equation as well. As a result, we were left with the option of building a shake table from scratch, one that can compete with superior alternatives costing tens of thousands of dollars. We went through many different ideas: one rotating motor, two rotating motors, one linear actuating motor with sliding rails, and even using one motor in two different directions and combining the two axes. After the plethora of efforts to design the table, we concluded that the shake table was not the best way, under our time constraints for different events (SEAS and VJAS), for creating precise, 2D seismic data, in the form that our prototype was designed to pick up on.

As an alternative method, we will be testing each sensor independently in their feasibility to work in the prototype further. For the accelerometer, we will be using a semi pendulum setup; the Accelerometer and Microcontroller sit coupled to a plywood plank handing from a point. The plywood should collide with a wall when dropped from differing angles, leaving a sudden drop in acceleration when it collides with the wall. We compare experimental data to theoretical calculations done by using energy conservation and kinematics. Calculated error tells us the accuracy of a cheap sensor in a high impulse event, alongside indicating any prototyping needed for the actual sensor.

For the Piezoelectric disc, we must solder them onto perf bored to allow for a clean connection and ease of prototyping with additional components. We will the couple it too with a high-density wall that contains saturated soil as the formula for acoustic Impedence is $Z= \rho (v)$, meaning higher density equates to a more precise shock to the PZT. Voltage from the sensor will

be collected and converted to analog via the ADS115 analog to digital converter. Due to possible hardware anomalies and constrained, it is intended that we observe results more qualitatively to retrieve results.

The physics of our project will not work however, if there is any internal shaking in the prototype. To remedy this, we CAD designed a special casing guarding the prototype:

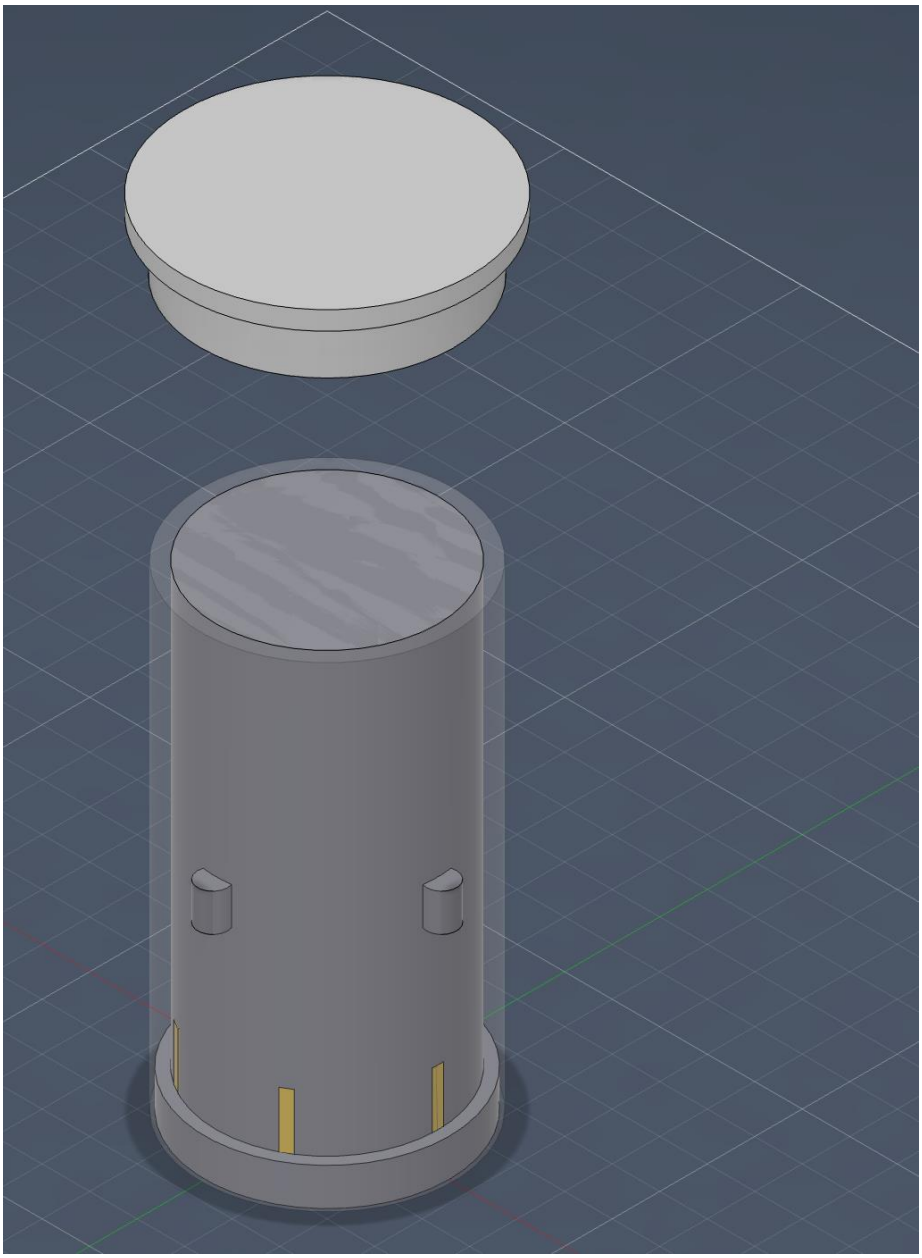

Figure 1: This view of the casing portrays the white lid of the casing hovering over the open casing. The transparent covering is the PVC pipe, which is the main cylindrical body, and is there to protect the prototype and inner contents of the casing from soil pressure and prevents the casing from corroding due to the moisture barrier in the soil and acids. The light grey monocoque frame is inside of the surrounding PVC pipe and serves as the main structural shell which bears stress loads and resists compression from the soil.

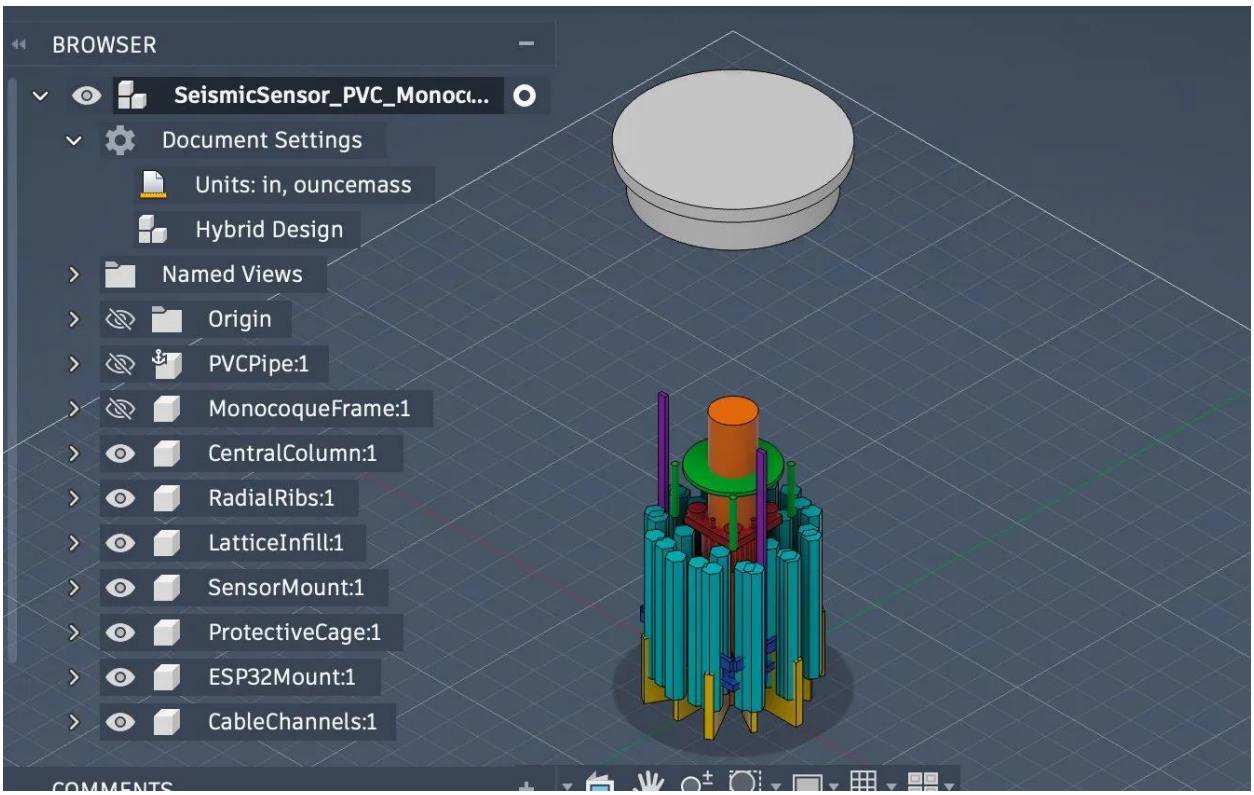

Figure 2: This view hides the PVC pipe and Monocoque Frame, showing the internal contents of the casing. First, the orange column in the middle is the central column which is the core spine of the assembly and transfers the different vertical loads, resisting buckling when the casing is buried. The yellow rib-like structure at the bottom is called the radial ribs, which prevents ovalization under the lateral soil squeeze. The cyan infill is called the lattice infill which basically serves as a shock absorbing filler and also a vibration damping layer. It also reduces

mass vs solid fill and uses the air gaps from this to act as a thermal buffer, allowing heat to escape.

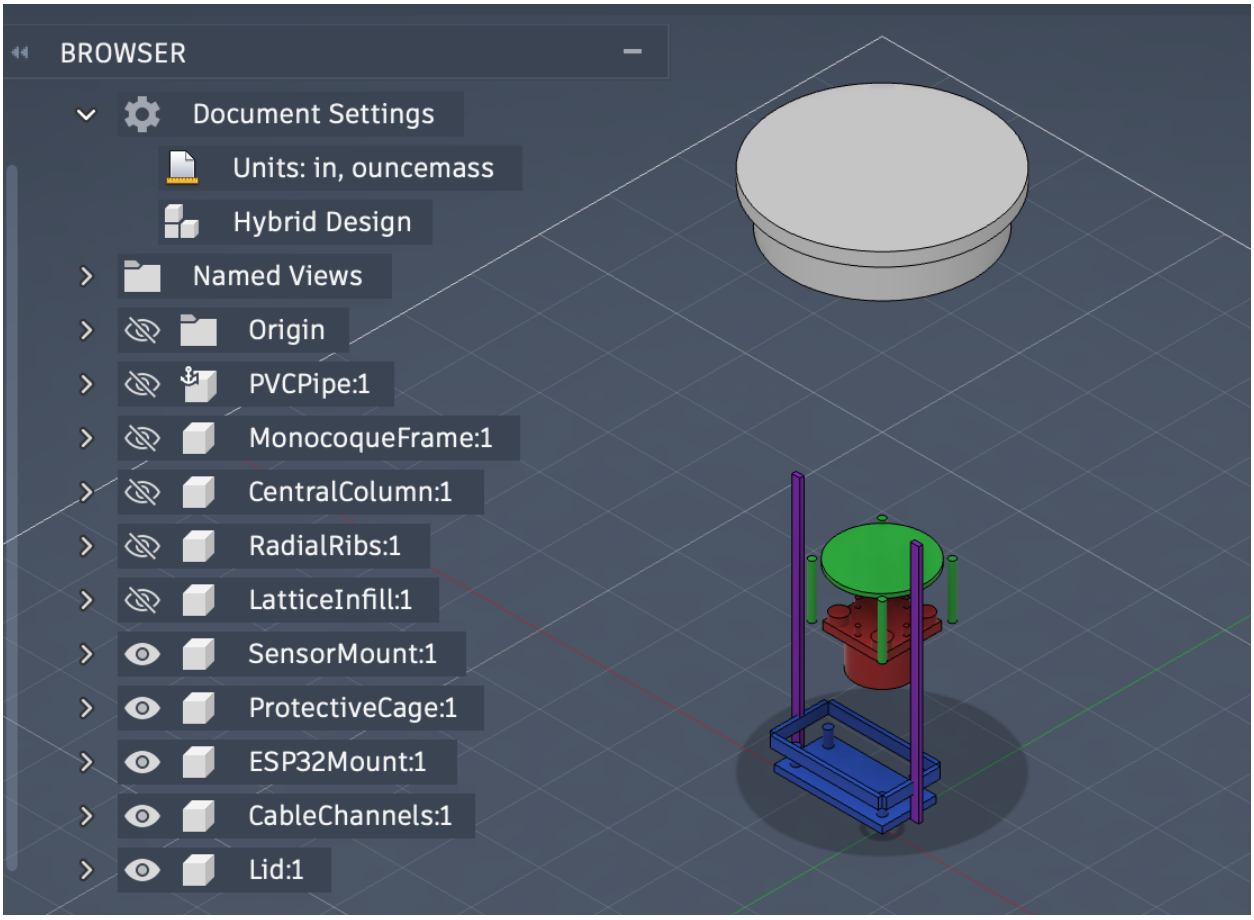

Figure 3: This view presents the protective cage, sensor mount, ESP32 microcontroller mount, cable channels, and hovering lid. The sensor mount holds the MEMS accelerometer sensor and couples the sensor to the casing. This mount prevents any rattling of the sensor itself. In green is the protective cage, in which guards the PCB and shields from intrusion. It absorbs impact at any point and has airspace to let heat escape. In blue is the ESP32 mount in which secures the microcontroller and uses standoffs to increase airflow. This mount also provides strain relief for General Purpose Input/Output (GPIO), which are pins on the edge of the ESP32 board in which read in a signal, or output a signal. In technicality, the GPIO is how the ESP32

can communicate with any alarm system attached to the casing. The cable channels are the vertical purple slots in which the wiring system is a controlled, stable path out of the sealed casing and into the surrounding soil.

Moving over the machine learning side, we fixed a big issue in the earlier pipeline where the model couldn't learn properly because of how the data was being inputted in. The acceleration and velocity signals were getting mixed together in the wrong way. As a result, it could not pick up how things changed over time across sensors. We solved the issue by reshaping the data in a clean (200,2) format, which finally let the model see the patterns correctly. Additionally, we improved how the data gets normalized. Before, tiny background vibrations were blown out of proportion. This made the model think a normal noise was an earthquake which resulted in an extremely high false alarm rate.

On the architecture side, we decided to move away from the simple CNN and built a structure way more capable: a multi-scale residual CNN with squeeze-and-excitation (SE) attention. In simpler terms, the model looks at the data at different zoom levels at the same time, using different kernel sizes (3,7,11) so it can catch both quick P-waves spikes and slower S-wave patterns. We realized that the SE blocks help the model decide what matters more depending on the moment. Specifically, the acceleration or velocity signal is more useful. We also trained the model more carefully by using balanced data from the 19 datasets and intentionally feeding it hard negative samples, like tricky noise cases, so it would not get fooled easily. The result of all these changes shows a huge jump in performance: 99% detection rate on the Dominican Republic test set, with only 6.8% false alarm rate. A major improvement from the previous 52% detection rate. We did notice that the 99% detection rate is usually high and could indicate overfitting, which would cause problems in future testing.

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